Example - 3

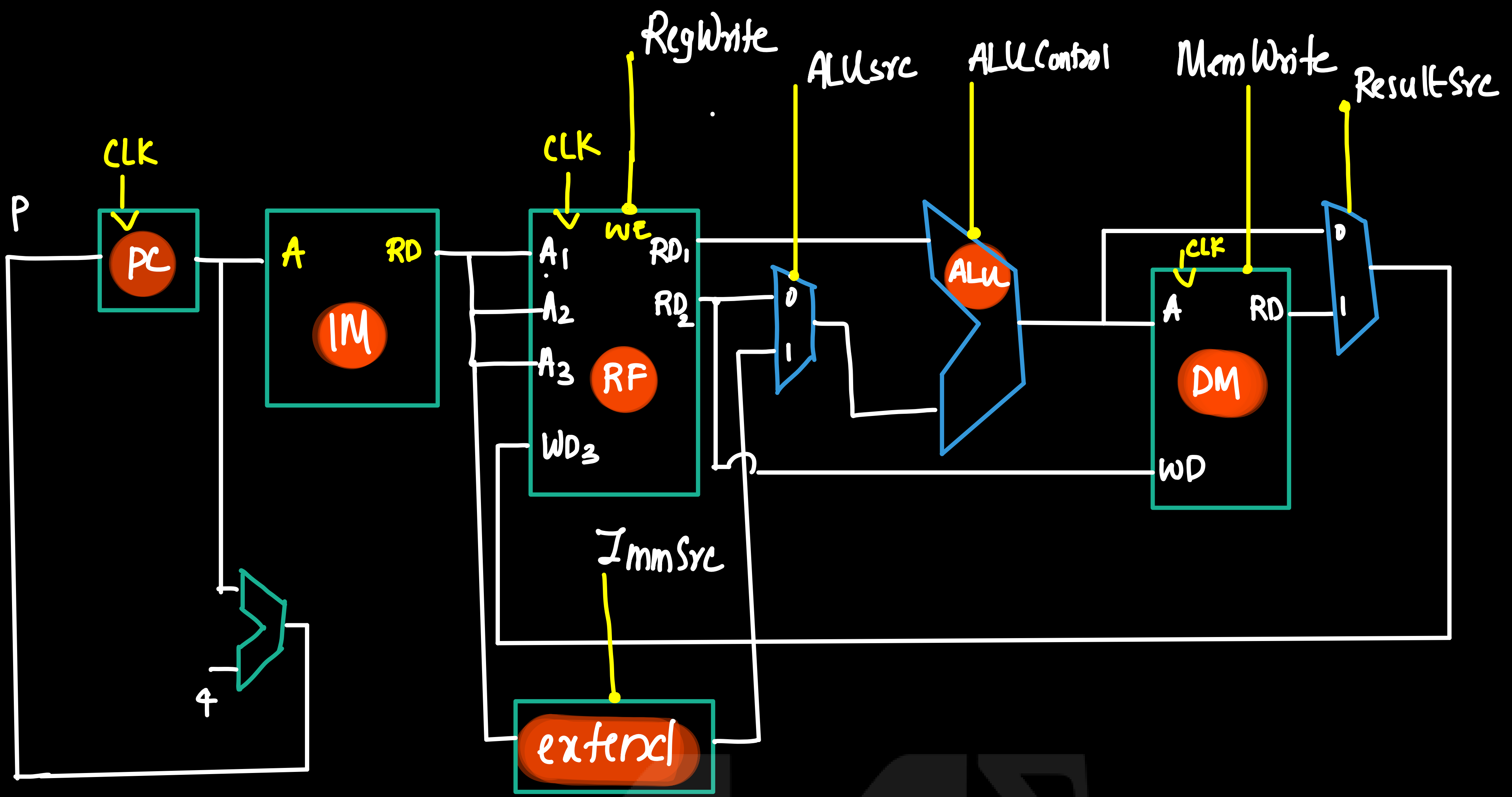
or x_4, x_5, x_6

- operation - perform OR operation b/w x_5, x_6 and store the result in Register x_4
- it is a R-type instruction

Instruction format

7	5	5	3	5	7
f_7	rs_2 (x_6)	rs_1 (x_5)	f_3 (110)	rd (x_4)	op
(31 : 25)	(24 : 20)	(19 : 15)	(14 : 12)	(11 : 7)	(6 : 0)
	00110	00101		00100	

Data path

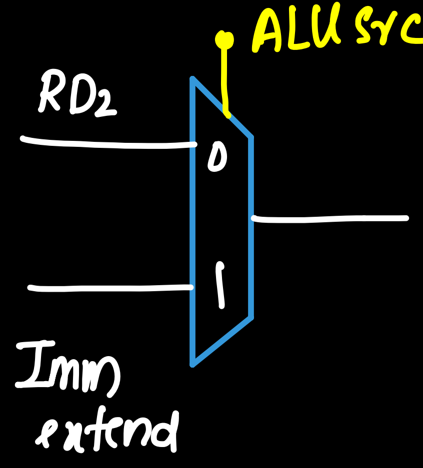


1. PC Contains address of next instruction to be executed
2. PC places address to the address input (A) of instruction memory.
3. instruction memory fetch the instruction and place in read data output (RD)
4. Instruction or x_4, x_5, x_6.
5. instruction bits representing x_5 (instn 19:15) connected to address input A_1 of register files.
6. instruction bits representing x_6 (instn 24:20) connected to A_2.
7. Register file read the content of x_5 and place the content in RD_1
8. Register file read the content of x_6 and place the content RD_2.
9. Data available at RD_1 is directly given to the one input of ALU
10. Data available at RD_2 and output of extender is connected to a multiplexer with a select line

(ALUSrc).

If ALU Src = 0, then multiplier select register file output
If ALU Src = 1, then select Extended immediate value.

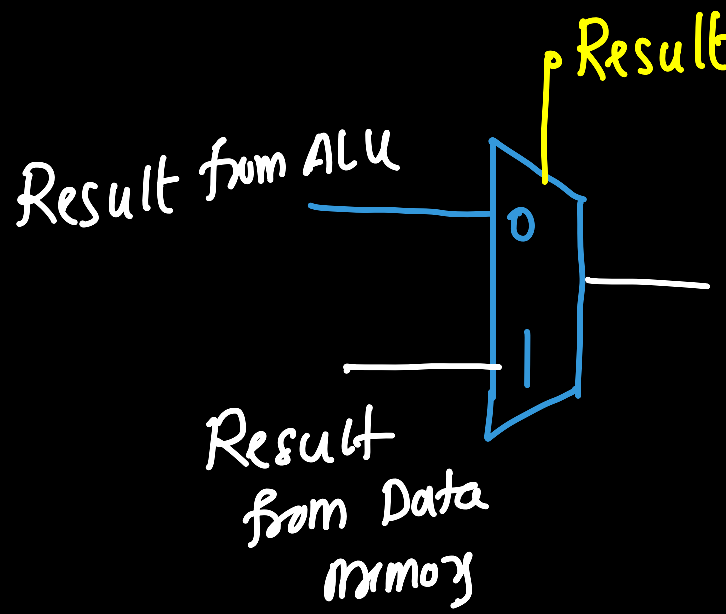
Here register file output (RD_2) is required so ALUSrc=0



(11) ALU perform the operation b/w registerfile outputs RD_1 & RD_2. Operation of ALU is decided by the ALU control signal.

ALU control signal for or operation is 011

(12) Result of ALU write back to register x_4. But for Lw/sw instructions result comes from data memory. For R type (eg: or s_4, s_5, s_6) result comes from ALU. So we use a multiplexer to chose the proper result based on the type of instruction.



If ResultSrc = 0, Result selected from ALU
If ResultSrc = 1, Result selected from Data memory.

In this example, we need result of ALU so ResultSrc=0

(13) write back the ALU result in register file (x_4)

(Connect instruction bits corresponding to x_4 to A_3 and make the control signal (RegWrite) to 1

↳ store the result in register x4.

N

Sample Questions – Single cycle Data path.

① Explain data path operation for the following instruction in a single cycle RISC processor with the help of a neat diagram.

1. lw x4, 5(x9)
2. sw x5, 5(x9)
3. add s4, s5, s6
4. and s4, s5, s6
5. beq s0, s1, L1